AGH University of Science and Technology

Trigonometric Fourier Series (part 2) Dirichlet cond and pointwise convergence for Electronics and Telecommunications

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Outline (Part 2)

- Orthogonal and orthonormal sequence and series (revision),
- Orthogonal system of rescaled sinuses and cosines (*harmonics*),
- Dirichlet condition and pointwise convergence,
- The Euler-Fourier Theorem for the complete system of harmonics.

Orthogonal Sequence of Functions (revison)

Orthogonal Sequence of Functions

The sequence of functions:

$$(\varphi_n(x)) = \varphi_0(x), \varphi_1(x), \varphi_2(x), \ldots$$

is called orthogonal on the interval $\langle a, b\rangle$ if:

(1) $\forall m, n \in \mathbb{N}, m \neq n : \langle \varphi_m, \varphi_n \rangle = 0$ (any two are orthogonal)

(2) $\forall : \|\varphi_n\|_2 > 0$ (no zero functions in this sequence)

Orthonormal sequence of functions (revision)

Orthonormal sequence of functions

Sequence of the functions $(\varphi_n(x))$ is called orthonormal if:

$$\langle \varphi_n, \varphi_m \rangle = \delta_{nm} = \begin{cases} 0 & \text{if } n \neq m \\ 1 & \text{if } n = m \end{cases}$$

(δ_{nm} is called Kronecker delta).

In other words, the sequence of functions is orthogonal and the norm of each of those functions is 1, then it is orthonormal.

Orthonormal sequence of trigonometric functions

We shall now check that the system of rescaled sinuses and cosines forms orthogonal system of functions on $\langle -\pi, \pi \rangle$.

Towards trigonometric Fourier series

Rescaled sinuses are othogonal

The sequence $(\varphi_n(x)) = (\sin(x),\ \sin(2x),\ \sin(3x),\ \ldots)$ is orthogonal on $\langle -\pi, \pi \rangle$.

because....

(1) for $m \neq n$: $\langle \varphi_m, \varphi_n \rangle = \int_{-\pi}^{\pi} \sin(nx) \cdot \sin(mx) \mathrm{dx}$
$= \frac{1}{2} \int_{-\pi}^{\pi} \cos[(n-m)x] \mathrm{dx} - \frac{1}{2} \int_{-\pi}^{\pi} \cos[(n+m)x] \mathrm{dx} = 0$

(2) for $n = m$: $\langle \varphi_n, \varphi_n \rangle = \int_{-\pi}^{\pi} \sin^2(nx) \mathrm{dx} = \left[\frac{x}{2} + \frac{1}{2n} \cdot \sin[2nx] \right] \Big|_{-\pi}^{\pi} = \pi$

Towards trigonometric Fourier series

Rescaled cosines are orthogonal

The sequence $(\varphi_n(x)) = (\cos(x),\ \cos(2x),\ \cos(3x),\ \ldots)$ is orthogonal on $\langle -\pi, \pi \rangle$.

because...

$\langle \varphi_m, \varphi_n \rangle = \int_{-\pi}^{\pi} \cos(nx) \cdot \cos(mx) \mathrm{dx}$
$= \frac{1}{2} \int_{-\pi}^{\pi} \cos[(n-m)x] \mathrm{dx} + \frac{1}{2} \int_{-\pi}^{\pi} \cos[(n+m)x] \mathrm{dx}$ etc..

Rescaled sinuses and cosines

...are like an orthogonal "base" in the infinitely dim space...

The orthogonal complete system of trigonomertic functions

The sequence $(\varphi_n(x)) = (1, \sin(x),\ \cos(x),\ \sin(2x),\ \cos(2x),\ \sin(3x),\ \cos(3x),\ \ldots)$ is orthogonal on $\langle -\pi, \pi\rangle$ and it forms an orthogonal complete* system of functions on $\langle -\pi, \pi\rangle$.

It is left to check that 1 is orthogonal to any other function above (easy), and that any sines is orthogonal to any cosine from

$$\int_{-\pi}^{\pi} \sin(nx)\cdot\cos(mx)\mathrm{dx} =$$
$$= \tfrac{1}{2}\int_{-\pi}^{\pi} \sin[(n-m)x]\mathrm{dx} + \tfrac{1}{2}\int_{-\pi}^{\pi} \sin[(n+m)x]\mathrm{dx}.$$

Orthogonal Trigonometric Series

Orthogonal Series

Let $(\varphi_n(x)) = (1, \sin(x),\ \cos(x),\ \sin(2x),\ \cos(2x),\ \sin(3x),\ \cos(3x),\ \ldots)$ be an orthogonal sequence on $\langle -\pi, \pi \rangle$ and (γ_n) be any number sequence then

$$\sum_{n=0}^{\infty} \gamma_n \varphi_n(x)$$

is an orthogonal series.

Convergence of an orthogonal trigonometric series

Lemma (Fourier trigonometric Thm)

Assume f is square integrable on $\langle -\pi, \pi \rangle$ and
$(\varphi_n(x)) = (1, \sin(x),\ \cos(x),\ \sin(2x),\ \cos(2x),\ \sin(3x),\ \cos(3x),\ \ldots)$
is an orthogonal complete system of functions on $\langle -\pi, \pi \rangle$.
If the following orthogonal series is convergent in the mean square norm on the interval $\langle -\pi, \pi \rangle$ to f ie.

$$d_2\Big(\sum_{n=0}^{N} \gamma_n \varphi_n, f\Big) = \int_a^b \Big(\sum_{n=0}^{N} \gamma_n \varphi_n(x) - f(x)\Big)^2 dx \underset{N \to +\infty}{\longrightarrow} 0,$$

then $\gamma_n = \frac{\langle f, \varphi_n \rangle}{\|\varphi_n\|_2^2}$.

Fourier trigonometric Thm in the specific settings

What follows in the next three slides is strictly speaking not called Euler-Fourier Theorem, but those formulas are repeated here to show you some similarity between an abstract theorem (previous slide, with mean square convergence convergence) and specific formulas in the trigonometric setting (next three slides, where there is only pointwise convergence under Dirichlet conditions).
Dirichlet conditions are explained in detail after those three specific theorems.

Fourier trigonometric Thm for 2π - periodic function

If a 2π - periodic function f satisfies the Dirichlet Conditions then it can be expanded at any point $x \in \langle -\pi, \pi \rangle$ to its Fourier series ie.

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos(nx) + b_n \sin(nx)]$$

where coefficients are given by the formulae

$$a_0 = \frac{1}{\pi} \int_{2\pi} f(x) dx \tag{1}$$

$$a_n = \frac{1}{\pi} \int_{2\pi} f(x) \cos nx dx, \; n \geq 1 \tag{2}$$

$$b_n = \frac{1}{\pi} \int_{2\pi} f(x) \sin nx dx, \; n \geq 1 \tag{3}$$

where integrations are over any interval in x of the length 2π.

Fourier trigonometric Thm for $2l$ - periodic function

If a $2l$ - periodic function f satisfies the Dirichlet Conditions then it can be expanded at any point $x \in \langle -l, l \rangle$ to its Fourier series ie.

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{+\infty} a_n \cos\frac{n\pi x}{l} + b_n \sin\frac{n\pi x}{l},$$

where coefficients are given by the formulae

$$a_0 = \frac{1}{l}\int_{-l}^{l} f(x)dx \tag{4}$$

$$a_n = \frac{1}{l}\int_{-l}^{l} f(x)\cos\frac{n\pi x}{l}dx, \;\; n \geq 1 \tag{5}$$

$$b_n = \frac{1}{l}\int_{-l}^{l} f(x)\sin\frac{n\pi x}{l}dx, \;\; n \geq 1. \tag{6}$$

Fourier trigonometric Thm for l - periodic function

If a l - periodic function f satisfies the Dirichlet Conditions then it can be expanded at any point $x \in \left\langle -\frac{l}{2}, \frac{l}{2} \right\rangle$ to its Fourier series ie.

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{+\infty} a_n \cos \frac{2\pi n x}{l} + b_n \sin \frac{2\pi n x}{l},$$

where coefficients are given by the formulae

$$a_0 = \frac{2}{l} \int_{\frac{-l}{2}}^{\frac{l}{2}} f(x) dx \tag{7}$$

$$a_n = \frac{2}{l} \int_{\frac{-l}{2}}^{\frac{l}{2}} f(x) \cos \frac{2\pi n x}{l} dx, \; n \geq 1 \tag{8}$$

$$b_n = \frac{2}{l} \int_{\frac{-l}{2}}^{\frac{l}{2}} f(x) \sin \frac{2\pi n x}{l} dx, \;\; n \geq 1. \tag{9}$$

The Dirichlet Conditions

The Dirichlet Conditions

Function f satisfies Dirichlet Conditions on $\langle a, b\rangle$ if

1. f is piecewise monotonic (ie. $\langle a, b\rangle$ can be divided into a finite number of subintervals on which f is monotonic);
2. f is continuous on $\langle a, b\rangle$ apart from at most a finite number of points, at which it has discontinuities of the first kind;
3. at each point $f(x_0) = \frac{1}{2}(\lim_{x \to x_0^+} f(x) + \lim_{x \to x_0^-} f(x))$;
4. at the ends $f(a) = f(b) = \frac{1}{2}(\lim_{x \to a^+} f(x) + \lim_{x \to b^-} f(x))$.

Imprecisely we may say, that if f has a finite number of finite jumps over some bounded interval, then it can be at any point approximated by its Fourier series.

Pointwise convergence from the Dirichlet Conditions

The Dirichlet Theorem

If a $2l$ - periodic function f satisfies the Dirichlet Conditions then it can be expanded at any point $x \in \langle -l, l \rangle$ to its trigonometric Fourier series

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{+\infty} a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l},$$

where

$$a_0 = \frac{1}{l} \int_{-l}^{l} f(x) \mathrm{dx};$$

$$a_n = \frac{1}{l} \int_{-l}^{l} f(x) \cos \frac{n\pi x}{l} \mathrm{dx}, \quad n \geq 1;$$

$$b_n = \frac{1}{l} \int_{-l}^{l} f(x) \sin \frac{n\pi x}{l} \mathrm{dx} \quad n \geq 1.$$

Violation of the Dirichlet Conditions

Example.
$f(x) = \sin x$ satisfies the condition of the piecewise monotonicity (when we subdivide into a finite number of intervals appropriately restricted domain), but $g(x) = \sin(1/x)$ does not, because it has infinitely many local extrema close to the origin, where ”it makes infinitely many oscillations”.